Chethan Ramakrishna Reddy

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INTEREST

- Mechatronics Engineering, Modeling, Simulation, Model-based control, System optimization, Controller optimization of Energy systems.

EXPERIENCE

PRODUCT DEVELOPMENT ENGINEER | Ford Motor Company

Jan 2022 – Present | SF Bay Area/Remote, USA

- xEV model based calibration optimization.
- Powertrain energy management focus.
- Software in the loop focus.

RESEARCH ASSISTANT - ENERGY MECHATRICS LAB | Michigan Technological University

May 2017 – Jan 2022 | Houghton, MI, USA

- Research on Model Predictive Control of (i) Building HVAC System with Solar Energy Integration, and (ii) Internal Combustion Engine with Waste Heat Recovery at the Energy Mechatronics Laboratory.

ENGINEERING INTERN | Constructis

Aug 2020 – Dec 2020 | MI/SC/Remote, USA

- System (Mechanical, Electrical, and Software) Definition, Design, and Analysis for the Roadway Kinetic Energy Recovery Pilot Product.

TEACHING ASSISTANT | Michigan Technological University

Fall 2017, '18, '19; Spring '18, '20 | Houghton, MI

- Lab instructor for courses focusing on (i) dynamics and control of mechanical systems, and (ii) introductory manufacturing processes. Both these consists mostly of senior undergraduate students in Mechanical engineering.

CONTROL ENGINEERING INTERN | Halla Mechatronics

Jan 2019 – May 2019 | Bay City, MI, USA

- Closed-Loop (Plant and Control) Model Development, Validation and Simulation of Electronic Controllers in Motor Controls Group.

SENIOR ENGINEER/ENGINEER | Robert Bosch India

Sep 2015 - Aug 2016/Aug 2013 – Sep 2015 | Bangalore, India

- Modeling & Simulation Expert in the System Engineering Group (Responsible for Hybrid and Battery Electric Vehicles).
- Plant model development, control model development, integration of models, and system simulation in modeling and simulation group.

EDUCATION

PHD IN MECHANICAL ENGINEERING - MAR 2022

Michigan Technological University, Houghton, MI

Research focus - Model Predictive Control of Energy Systems for Heat and Power Applications

CGPA: 3.77/4

MTECH (MS EQUIVALENT) IN MECHATRONICS ENGINEERING - MAY 2013

National Institute of Technology Karnataka, India

CGPA: 8.37 / 10, US equivalent CGPA: 4 / 4

SKILLS

- Model Based System Engineering.
- Modeling, Simulation, App Design, Data Analysis, and Code Generation in Matlab/Simulink.
- Modeling, Simulation, and Data Analysis in Python
- System Know how/Modeling of thermal, mechanical, hydraulic, pneumatic, electronic and electric sub-systems. 0D, 1D, 3D Modeling Techniques.
- System Simulation (GT Suite, AVL, CarSim), Co-Simulation (CS) and Model Exchange (ME). FMI/FMU Standard.
- Model Based Calibration, Model Based Calibration Optimization.
- Numerical Optimization Techniques / CPU usage Optimization Techniques / Simulation Model Optimization Techniques. High Performance Computing, Parallel Processing.
- Model Based Optimal Control Design, Model Based Predictive Control Design. Linear & Non-Linear Model Based Optimization.
- Optimization Solvers & Techniques. Linear & Non-Linear Control Theory. Machine learning & Genetic Algorithms.
- Model, Software, and Hardware in Loop (MiL, SiL and HiL) Model Development, Integration, Validation & Verification.
- Mechanical CAD.
- ECU prototyping/HiL testing (Motohawk, dSPACE).
- Scripting/Software languages - Matlab (Read & Write), Python (Read & Write), C/C++ (Read).

PROJECTS

INDUSTRIAL

- Model verification
- Model based calibration optimization
- Simulation (of the digital twin) speed improvement
- Model-based Design, Testing, and Calibration.
- Modeling and Simulation of Electrically Assisted Power Steering System.
- Mechanical Design, Fabrication, and Controller Prototyping of Automotive Exhaust Active Noise Cancellation and Enhancement System.
- Hybrid Vehicle System Simulation.

ACADEMIC

PhD Thesis

- Model Predictive Control of Energy Systems for Heat and Power Applications

PhD Course Projects

- Fuel Consumption Reduction Technologies and Hybrid Vehicle Design
- Control System Development for a Hybrid Vehicle ECU (MotoHawk)
- Effect of External Supercharging in a CI Diesel Engine with Swirl Combustion Chamber (Simulation study)
- Efficacy of PV Solar Energy in Houghton, MI
- Optimal Control of Wave Energy Converters

Masters Thesis

- (Model-Based) Development of Automotive Thermo-Electric Generator (ATEG)

Bachelors Thesis '

- Design and Fabrication of Boundary Layer Turbine as a Potential Automotive Engine (Compressed Air as Fuel)

SELECTED PUBLICATIONS

- **C. R. Reddy** , V. Bonfochi Vinhaes, J. D. Naber, R. D. Robinett III, M. Shahbakhti, **"Model predictive control of a dual fuel engine integrated with waste heat recovery used for electric power in buildings"** , in *Optimal Control Applications and Methods*, 2022.
- M. Toub, **C. R. Reddy** , M. Razmara, M. Shahbakhti, R. D. Robinett, G. Aniba, **"Model-based predictive control for optimal MicroCSP operation integrated with building HVAC systems"** , in *Energy Conversion and Management*, Volume 199, 2019.